

ESP32 Flash Memory (4 MB)

NVS
0x9000
20 KB

OTADATA
0xE000
8 KB

APP0
0x10000
1.5 MB

SPIFFS
0x290000
1.5 MB

Security
Configuration

OTA
Firmware Update

Application
Firmware

Offline Data
Logging

AES-128 Key
Node ID
Temperature Thresholds

Boot Information
OTA Control

FreeRTOS Kernel
Edge Processing Tasks

CSV/TXT Files
Sensor History
Outage Buffer